

MILK RIVER, AB, Canada

WMO: 712440

Lat: 49.134N

Lon: 112.051W

Elev: 1050

StdP: 89.33

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.9	(c) -25.2	(d) -32.5	(e) 0.2	(f) -28.8	(g) -28.5	(h) 0.3	(i) -25.1	(j) 16.2	(k) 3.8	(l) 14.4	(m) 1.8	(n) 3.3	(o) 300	(p) 0.728

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 16.6	(c) 31.8	(d) 15.6	(e) 29.9	(f) 15.3	(g) 28.0	(h) 15.0	(i) 17.8	(j) 26.6	(k) 16.8	(l) 26.0	(m) 16.0	(n) 25.1	(o) 4.9	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.0	12.1	20.1	13.8	11.2	19.2	12.6	10.3	18.3	54.3	26.5	51.1	25.9	48.4	25.1	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 13.3	(b) 11.6	(c) 10.2	(e) -34.1	(f) 35.5	(g) 3.7	(h) 1.9	(i) -36.8	(j) 36.8	(k) -38.9	(l) 37.9	(m) -41.0	(n) 39.0	(o) -43.6	(p) 40.3		
(4) 13.3	(5) 11.6	(6) 10.2	(7) -34.2	(8) 20.6	(9) 3.7	(10) 3.1	(11) -36.9	(12) 22.8	(13) -39.0	(14) 24.6	(15) -41.1	(16) 26.3	(17) -43.8	(18) 28.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.2	-6.7	-5.5	-1.0	4.7	9.9	14.1	18.3	17.5	12.3	5.5	-1.3	-6.3
	DBStd	10.78	8.87	8.06	7.60	4.96	4.38	3.42	3.23	3.46	4.60	5.37	7.53	8.19
	HDD10.0	2552	517	433	341	167	55	6	1	1	27	156	342	507
	HDD18.3	4880	776	666	599	408	261	133	41	57	185	399	589	765
	CDD10.0	795	0	0	1	8	53	128	256	233	98	15	2	0
	CDD18.3	82	0	0	0	0	1	6	39	31	6	0	0	0
	CDH23.3	1963	0	0	0	4	63	154	797	722	214	9	0	0
	CDH26.7	692	0	0	0	0	9	34	304	276	67	1	0	0
Wind	WSAvg	4.6	4.9	4.7	5.0	5.1	4.9	4.5	4.0	3.8	4.2	4.8	4.8	4.8
Precipitation	PrecAvg	337	9	8	15	32	49	83	29	33	36	18	11	9
	PrecMax	512	17	25	43	61	99	271	60	86	99	49	29	20
	PrecMin	187	3	2	4	10	11	12	2	4	5	2	2	2
	PrecStd	82	4	6	9	19	26	59	20	21	23	13	6	4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.5	13.4	17.7	23.6	27.9	30.6	34.9	34.0	31.6	24.5	18.2	11.8
		MCWB	5.4	5.5	7.8	10.7	14.0	17.0	16.3	15.7	14.4	12.0	8.8	5.0
	2%	DB	8.2	9.5	14.7	20.2	25.1	27.2	32.1	31.8	28.7	20.7	14.2	8.2
		MCWB	3.8	3.7	6.5	9.4	12.7	15.0	16.2	15.2	13.6	10.5	7.0	3.5
	5%	DB	6.2	7.1	12.1	17.0	22.3	25.0	30.1	29.8	25.8	17.5	11.3	6.2
		MCWB	2.5	2.5	5.3	7.7	11.5	14.3	16.0	14.9	12.9	9.0	5.3	2.2
	10%	DB	4.6	4.9	9.3	14.2	19.6	22.7	28.0	27.7	22.6	14.6	8.6	4.3
		MCWB	1.5	1.1	3.8	6.3	10.4	13.5	15.7	14.6	11.8	7.4	3.8	1.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	6.3	8.7	11.4	14.9	18.5	19.9	18.0	16.0	12.9	9.7	5.7
		MCDB	10.1	12.5	16.2	21.0	24.9	26.4	28.5	27.0	25.9	22.4	16.5	10.3
	2%	WB	4.0	4.2	7.0	9.7	13.4	16.7	18.3	17.0	14.6	10.9	7.3	3.8
		MCDB	7.7	8.9	13.6	19.2	22.8	24.4	26.9	26.9	25.1	19.1	13.6	7.8
	5%	WB	2.8	2.7	5.6	8.2	12.3	15.4	17.3	16.1	13.4	9.4	5.6	2.4
		MCDB	5.9	6.8	11.4	16.1	20.6	22.7	26.3	26.1	23.8	16.6	10.8	5.9
	10%	WB	1.6	1.4	4.1	6.7	11.1	14.3	16.4	15.3	12.3	7.9	4.1	1.1
		MCDB	4.5	4.7	9.0	13.3	18.5	21.1	25.6	25.1	21.9	14.1	8.2	4.2
Mean Daily Temperature Range	5% DB	MDBR	11.7	12.0	12.4	13.4	13.9	13.8	16.6	17.0	15.7	14.1	11.9	11.6
		MCDBR	11.9	14.0	16.4	18.9	19.0	18.0	20.4	21.5	21.5	18.5	14.9	12.6
		MCWBR	8.3	9.0	9.4	9.5	8.6	8.1	7.7	8.0	9.0	9.7	8.6	8.6
	5% WB	MCDBR	11.8	13.5	15.7	17.5	16.8	15.8	17.4	18.0	19.1	17.1	14.2	12.3
		MCWBR	8.7	9.1	9.4	9.2	8.0	7.9	7.7	7.6	8.6	9.3	8.6	8.7
Clear-Sky Solar Irradiance	taub	0.252	0.260	0.279	0.317	0.331	0.331	0.333	0.351	0.313	0.282	0.255	0.242	
	taud	2.468	2.494	2.498	2.396	2.402	2.453	2.456	2.378	2.456	2.529	2.511	2.453	
	Ebn at Noon	838	915	947	932	928	928	920	886	892	873	828	801	
	Edh at Noon	59	73	87	108	112	108	106	109	91	70	56	52	
All-Sky Solar Radiation	RadAvg	1.26	2.25	3.50	4.84	5.76	6.44	7.11	5.71	4.07	2.52	1.42	1.00	
	RadStd	0.10	0.18	0.29	0.31	0.48	0.45	0.39	0.36	0.33	0.24	0.10	0.09	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page